

```

scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
GHFFfitfile = fullfile(scuts_path,'e800GPffCuts.mat');
if ~exist('GHFFfitData','var')
    ld = load(GHFFfitfile);
    GHFFfitData = ld.fit_src_struct;
end

```

```

mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);

cut_names = fieldnames(GHFFfitData);
for i=1:numel(cut_names)
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_instrument(mi);
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_sample(sample);
end

```

```

dir_name = "GP";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

if ~exist('fit_reEi800GP','var')
    res_name = 'FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat';
    if isfile(res_name)
        ld = load(res_name);
        fit_reEi800GP = ld.fit_reEi800GP;
    else
        fit_reEi800GP = struct();
    end
end

for i=1:numel(cut_names)
    acolor k
    fprintf('*****\n')
    fprintf('**** CUT: %s\n',cut_names{i});
    fprintf('*****\n')

    cut_to_work = GHFFfitData.(cut_names{i});
    if isfield(fit_reEi800GP,cut_names{i})
        continue;
    end
    plot(cut_to_work); keep_figure;
    all_en = cut_to_work.data.p{2};
    en_range = min_max(all_en);
    cut_en = 100+half_dE:10:en_range(2)-half_dE;

```

```

[fr_800const_bg,res_name] = fit_multicuts_along_direction(...
    {cut_to_work},'FitEn_Ei800ff_1D',dir_name, ...
    1,cut_en,dE_step,half_dE,true);
fit_reEi800GP.(cut_names{i}) = fr_800const_bg;

acolor g
bg_data = extract_bg_par(fr_800const_bg.all_fit_par,[1,2]);
% original slope parameters

pd(bg_data(1));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure; ly 0 3;

pd(bg_data(2));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure

save(res_name,'fit_reEi800GP');
end

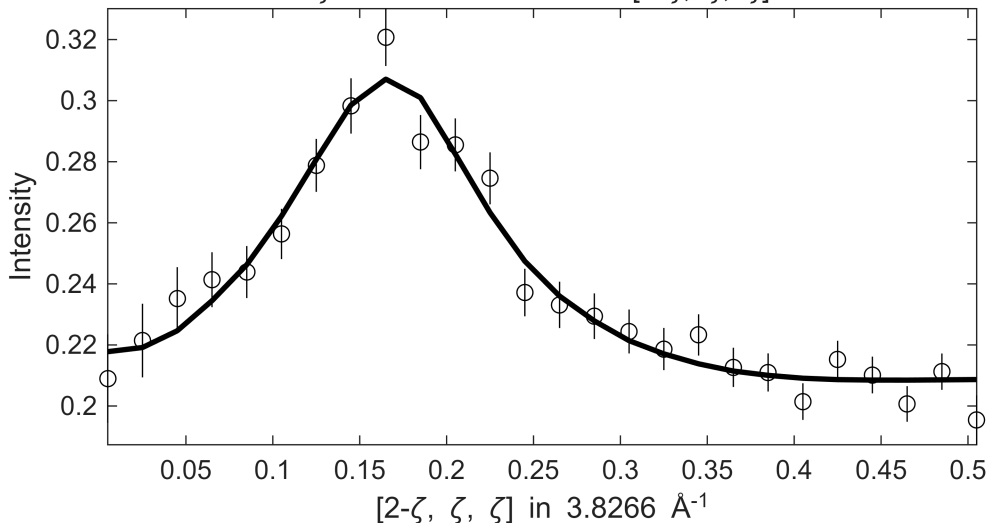
```

```

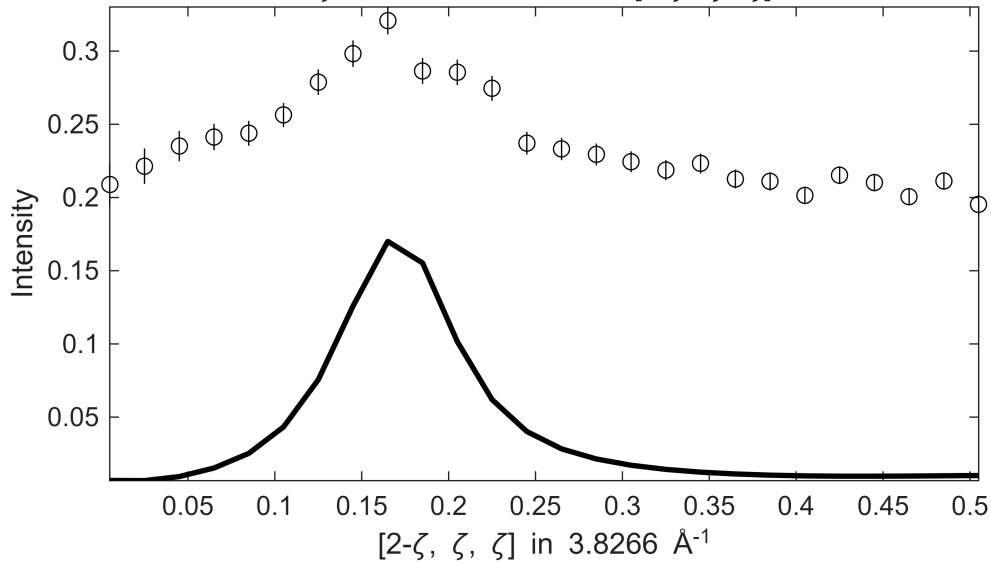
*****
**** CUT: Ei800_off110minFFGP
*****
**** CUT: Ei800_off200maxFFGP
*****
**** CUT: Ei800_off200minFFGP
*****
**** En = 105±5 going to 295
*****

```

Fe_ei800, w2_800MinFF.sqw
 S=0.75±0.16; J0= 34± 1.9; gamma=37.6289±13.2399
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $100 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

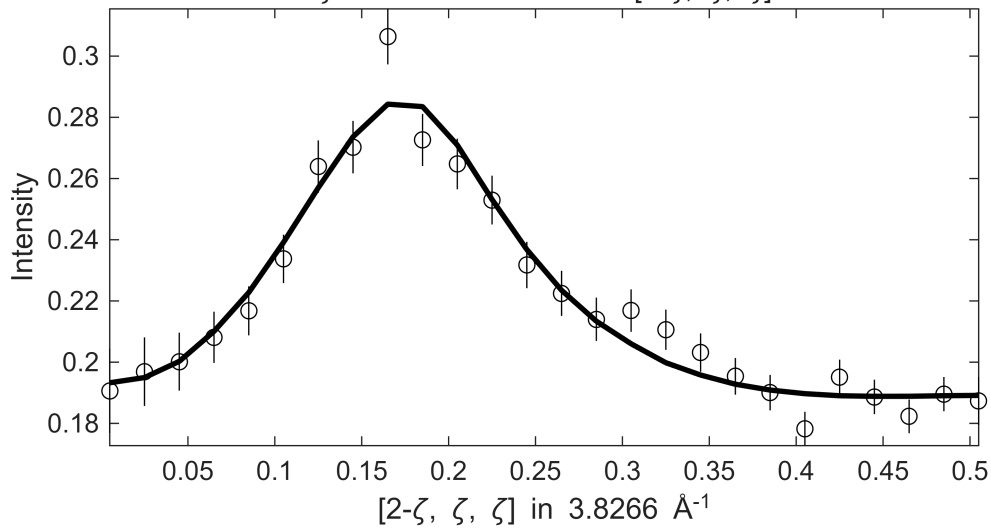


Fe_ei800, w2_800MinFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $100 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

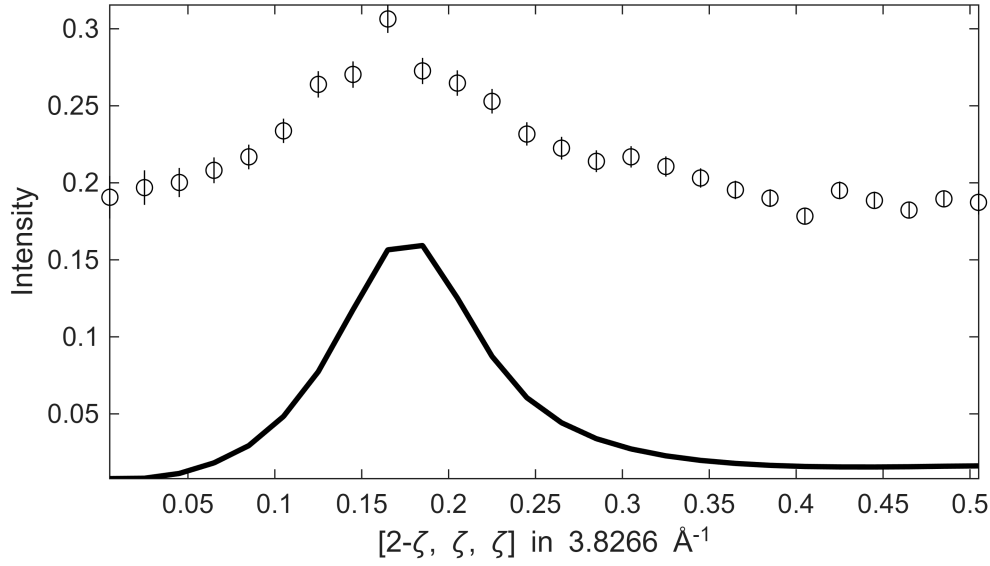


 **** En = 115±5 going to 295

Fe_ei800, w2_800MinFF.sqw
 $S=0.95 \pm 0.28$; $J_0 = 36 \pm 3$; $\gamma = 53.7032 \pm 21.5339$
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $110 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

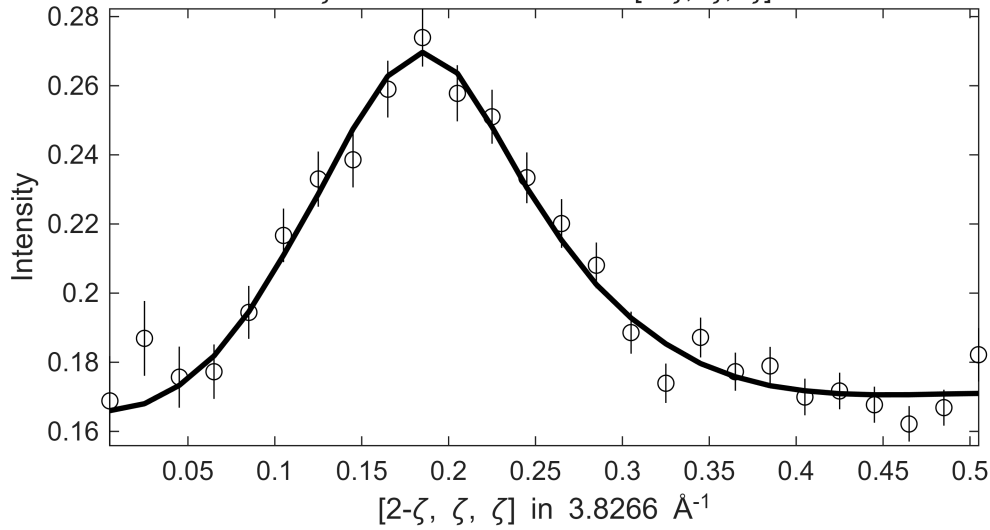


Fe_ei800, w2_800MinFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $110 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

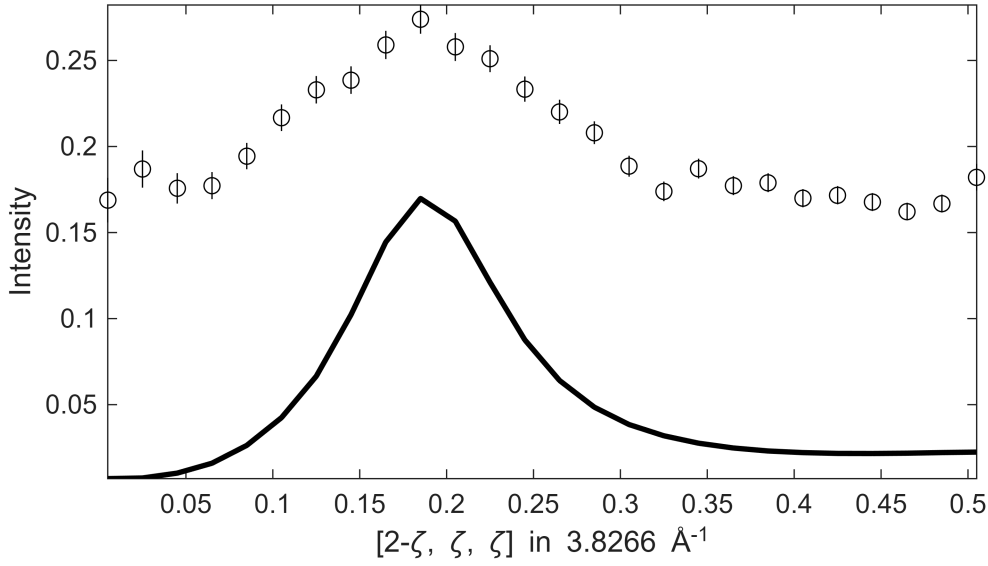


 **** En = 125±5 going to 295

Fe_ei800, w2_800MinFF.sqw
 $S = 1.1 \pm 0.033$; $J_0 = 37 \pm 0.53$; $\gamma = 59.001 \pm 1.33076$
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $120 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

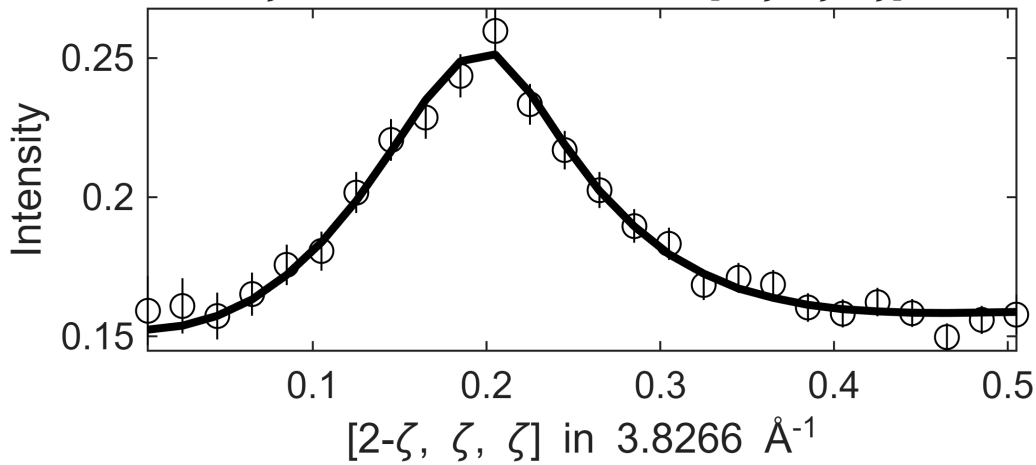


Fe_ei800, w2_800MinFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $120 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



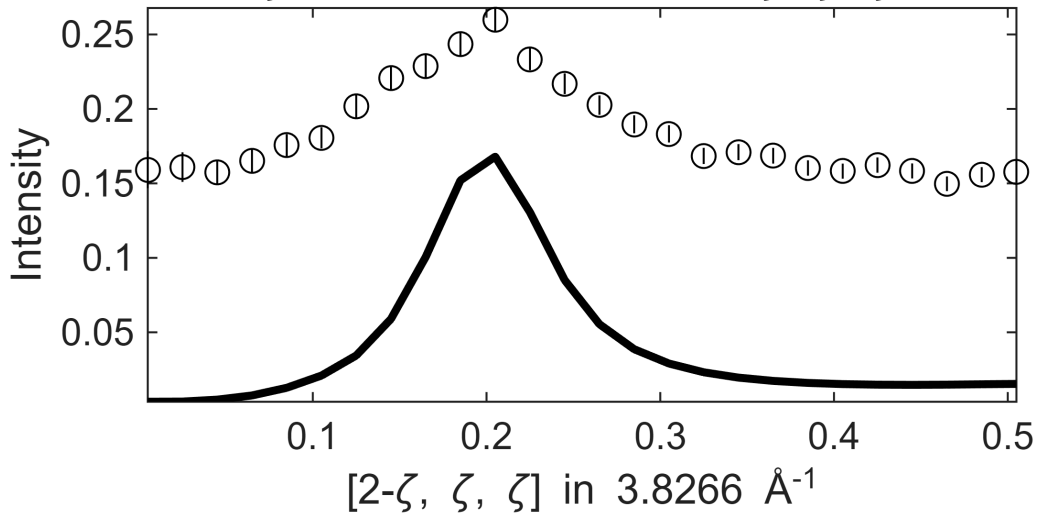
 **** En = 135±5 going to 295

Fe_ei800, w2_800MinFF.sqw
 S= 0.7 ± 0.016 ; J0= 34 ± 0.34 ; gamma= 38.2563 ± 0.732045
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

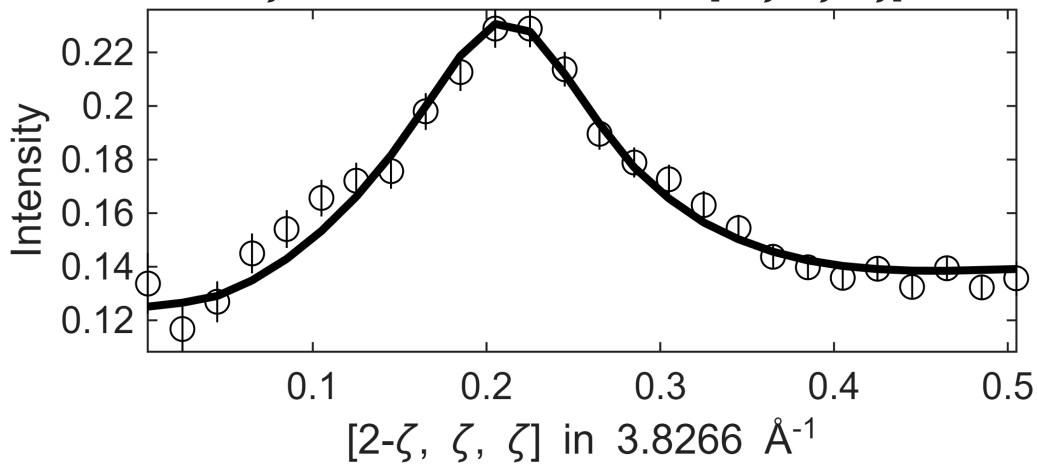
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 145±5 going to 295

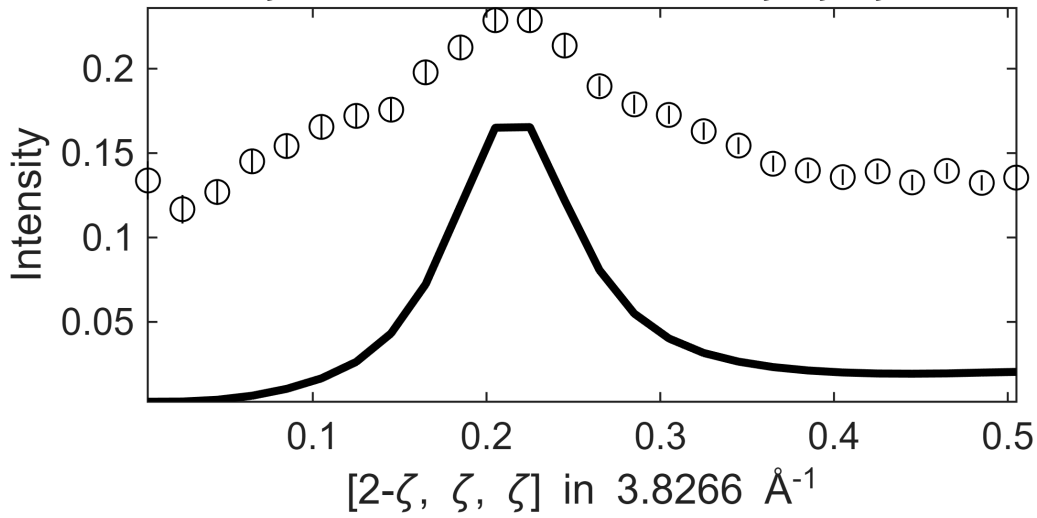
Fe_ei800, w2_800MinFF.sqw

S=0.71±0.025; J0= 34±0.39; gamma=39.2215±0.941456
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

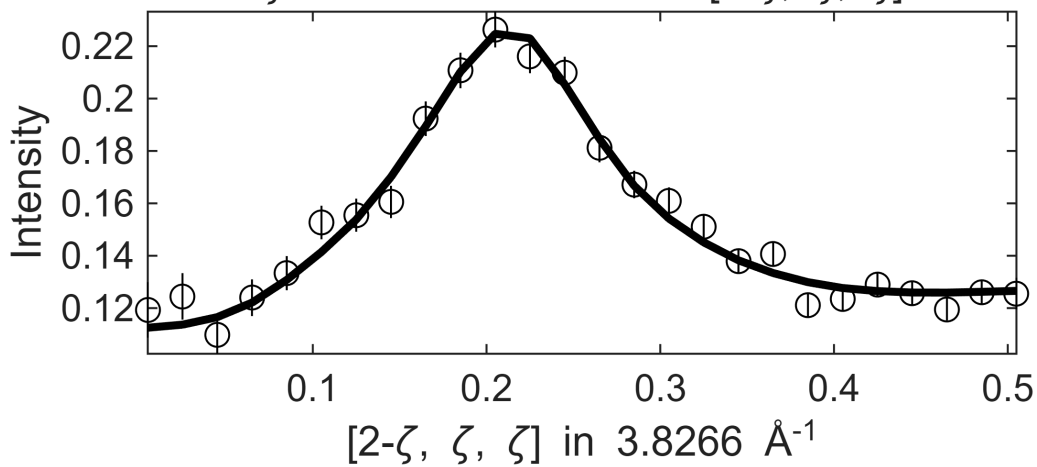
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



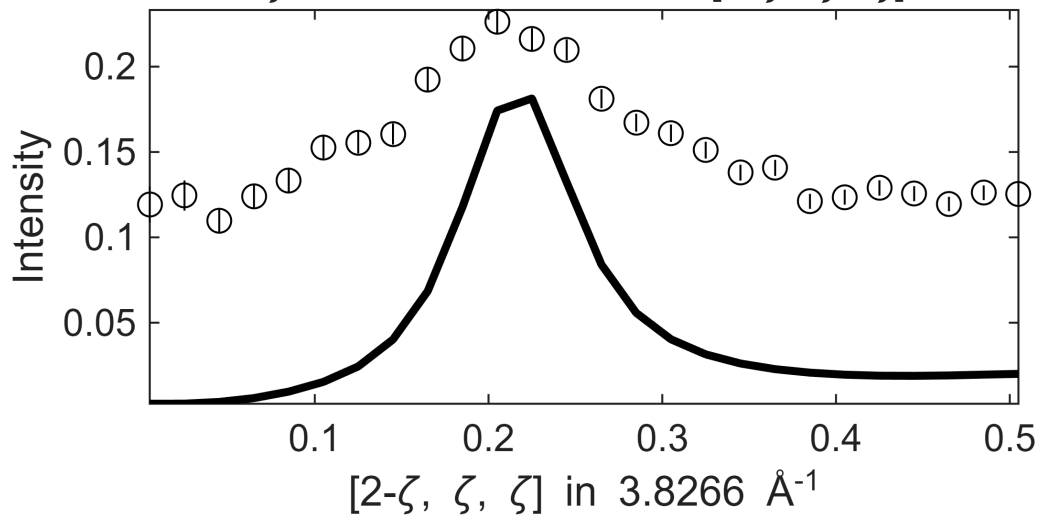
 **** En = 155±5 going to 295

Fe_ei800, w2_800MinFF.sqw

S=0.74± 0.1; J0= 35± 0.8; gamma=37.3143±7.51749
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

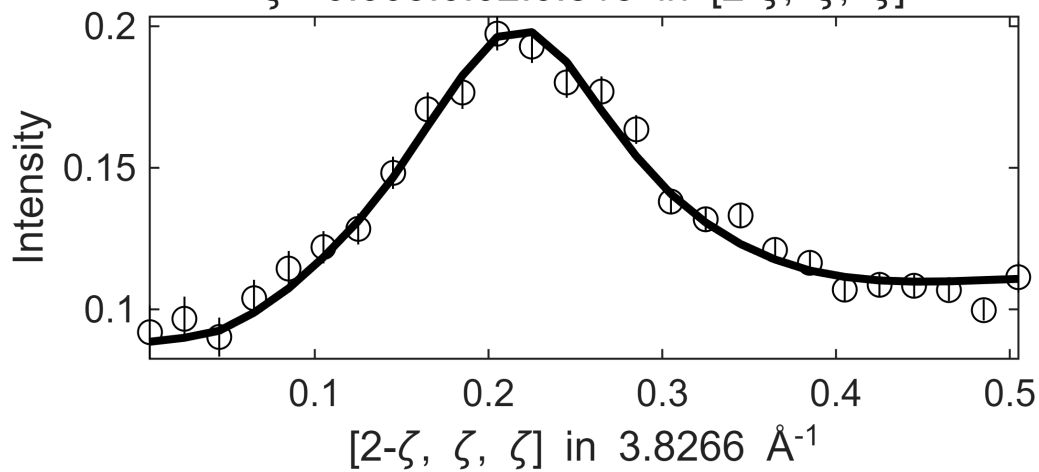


Fe_ei800, w2_800MinFF.sqw
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

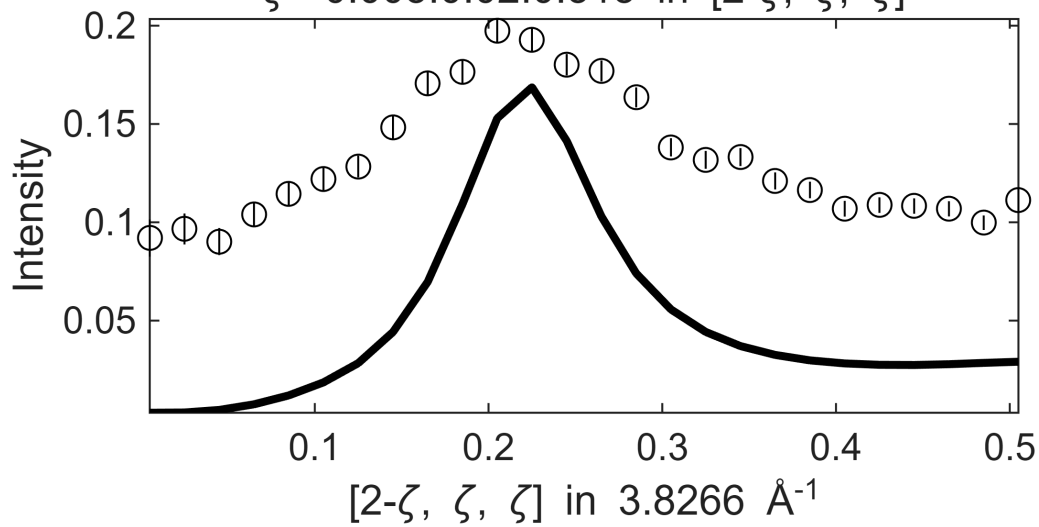


 **** En = 165 ± 5 going to 295

Fe_ei800, w2_800MinFF.sqw
 $S=0.89 \pm 0.031$; $J_0= 37 \pm 0.34$; $\gamma=51.176 \pm 1.12579$
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

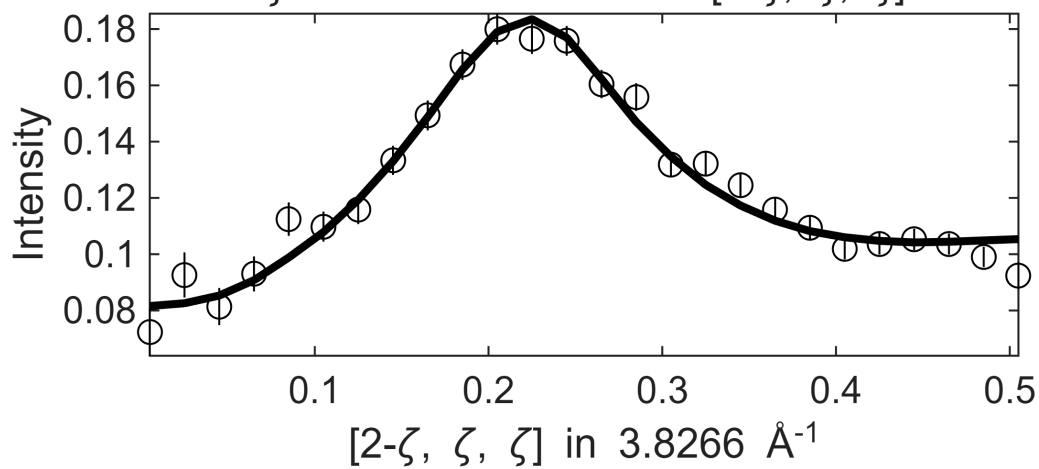


Fe_ei800, w2_800MinFF.sqw
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



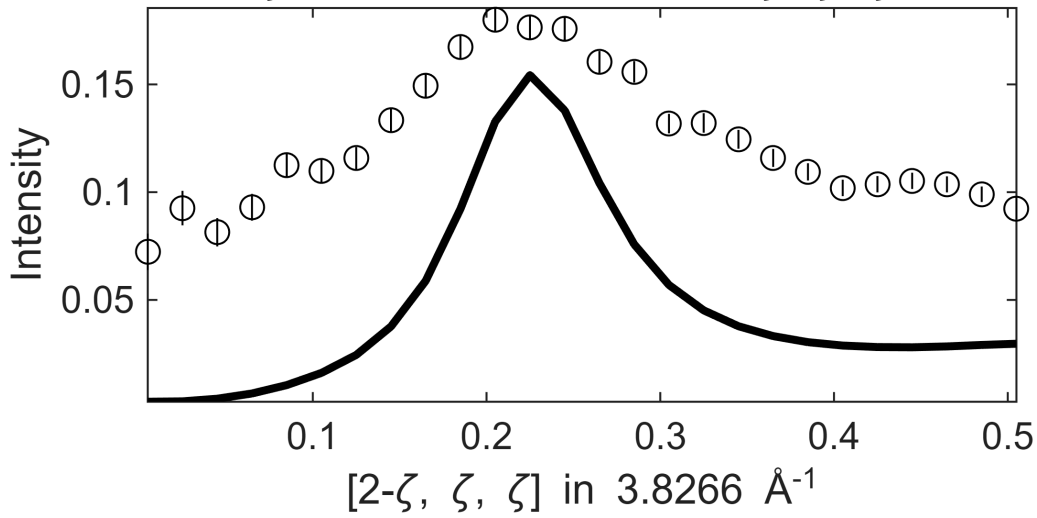
 **** En = 175±5 going to 295

Fe_ei800, w2_800MinFF.sqw
 S=0.84±0.033; J0= 38±0.36; gamma=53.5808±1.17887
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

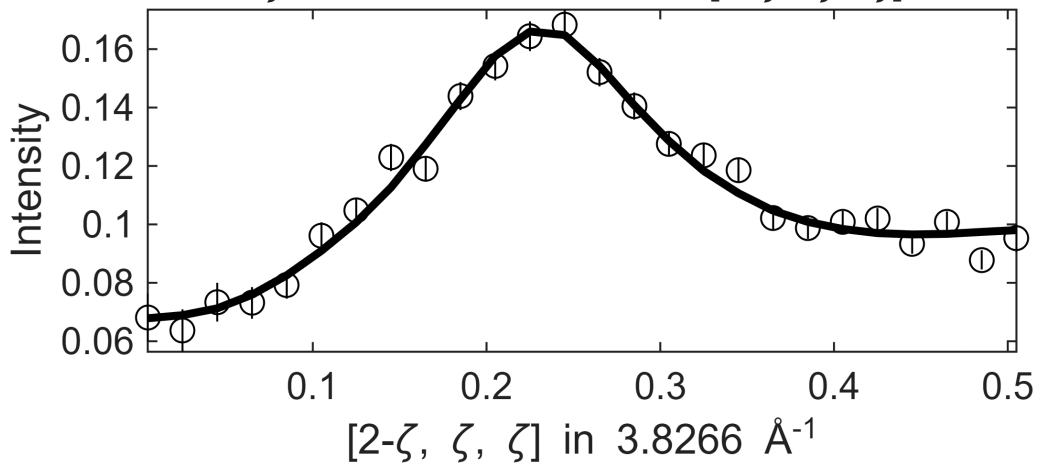


 **** En = 185±5 going to 295

Fe_ei800, w2_800MinFF.sqw

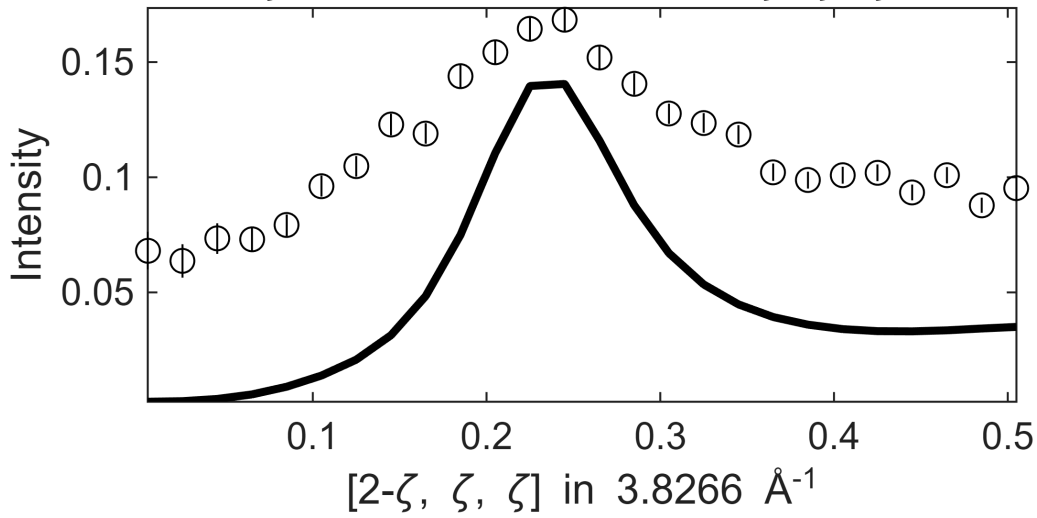
S=0.84±0.034; J0= 39±0.28; gamma=57.7792±1.24108

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

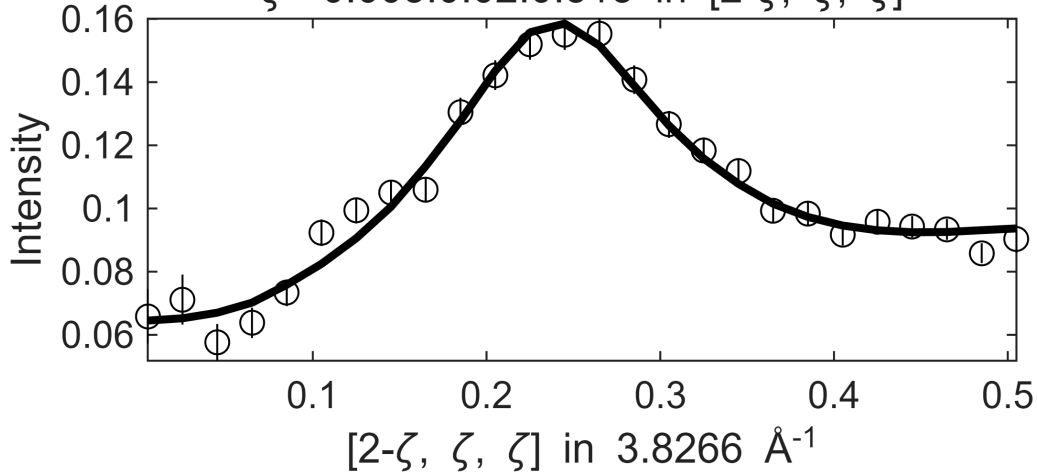


 **** En = 195±5 going to 295

Fe_ei800, w2_800MinFF.sqw

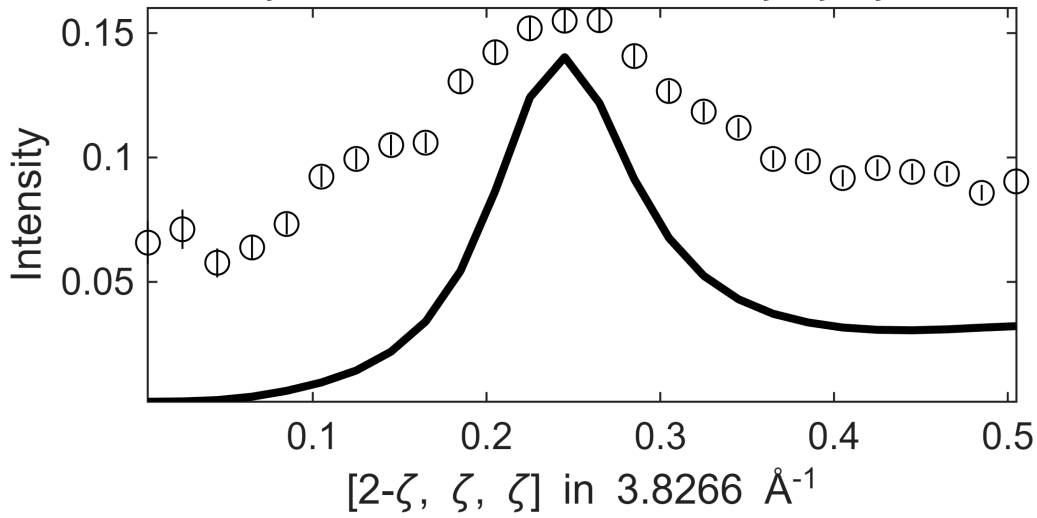
S=0.71±0.032; J0= 38±0.29; gamma=49.9935±1.03886

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

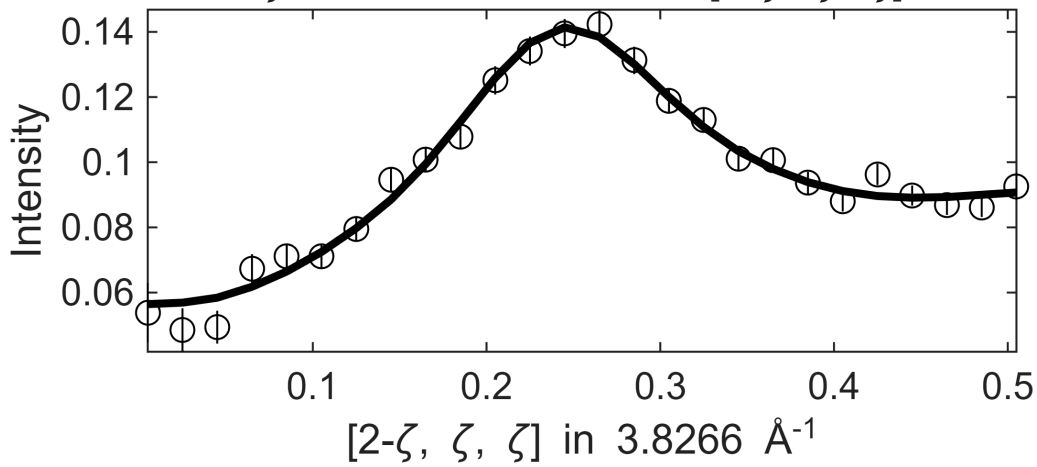
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 205±5 going to 295

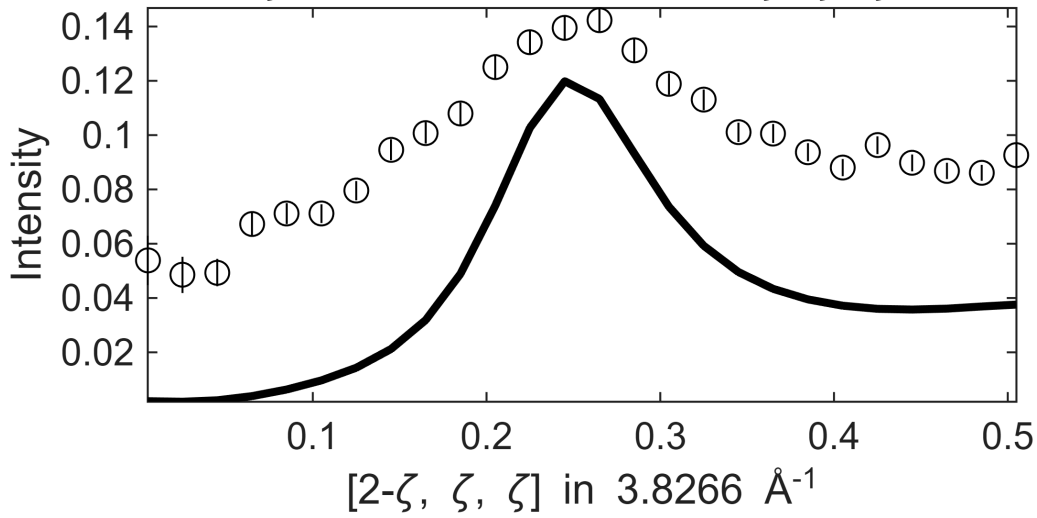
Fe_ei800, w2_800MinFF.sqw

$S=0.73\pm0.027$; $J_0= 40\pm0.21$; $\gamma=61.8831\pm1.04164$
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

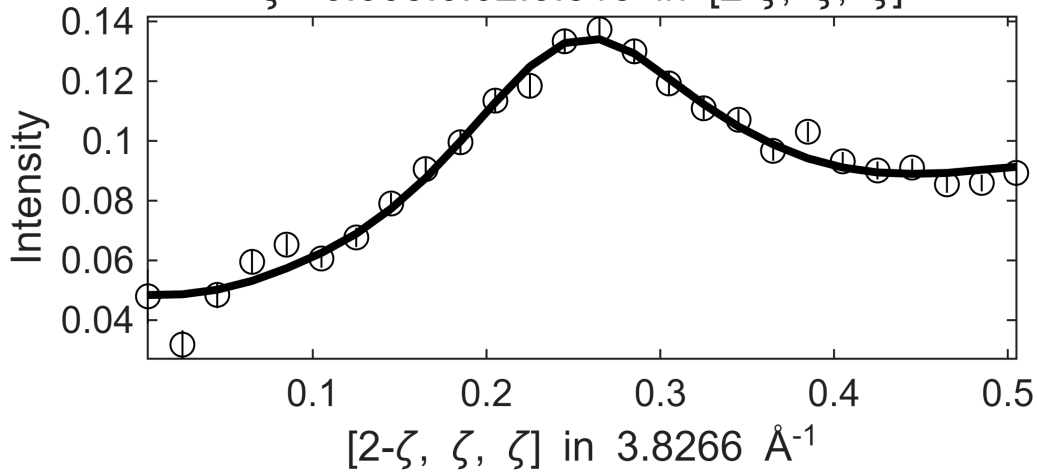


 **** En = 215±5 going to 295

Fe_ei800, w2_800MinFF.sqw

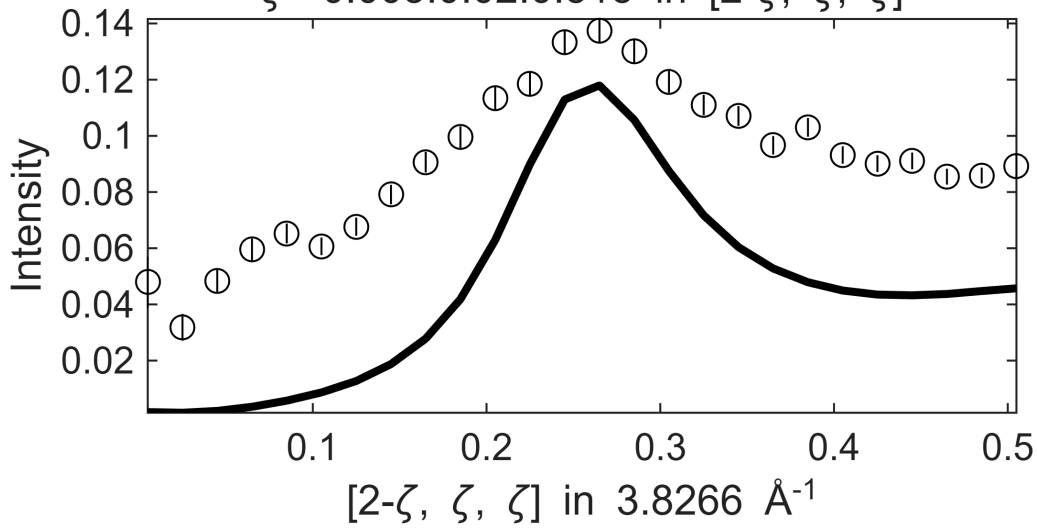
$S=0.73\pm0.033$; $J_0= 39\pm0.23$; $\gamma=64.1978\pm1.50032$

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

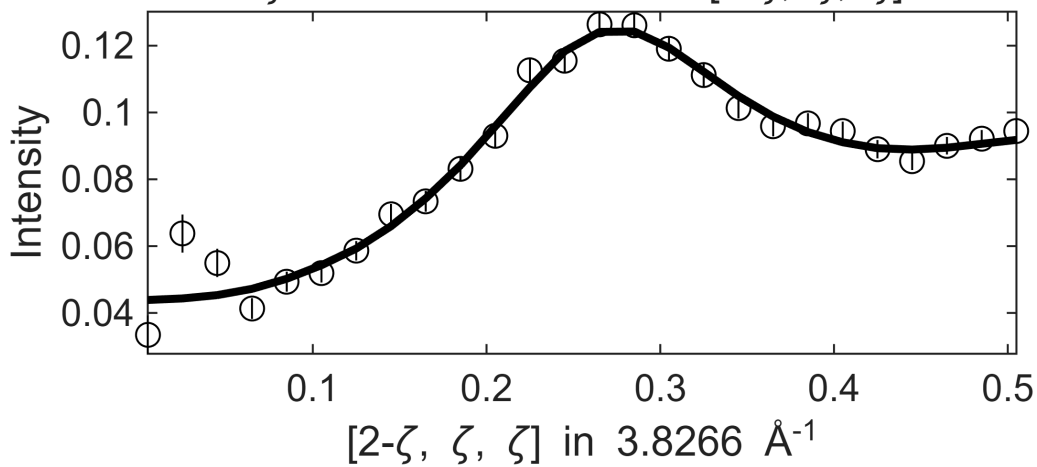
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 225±5 going to 295

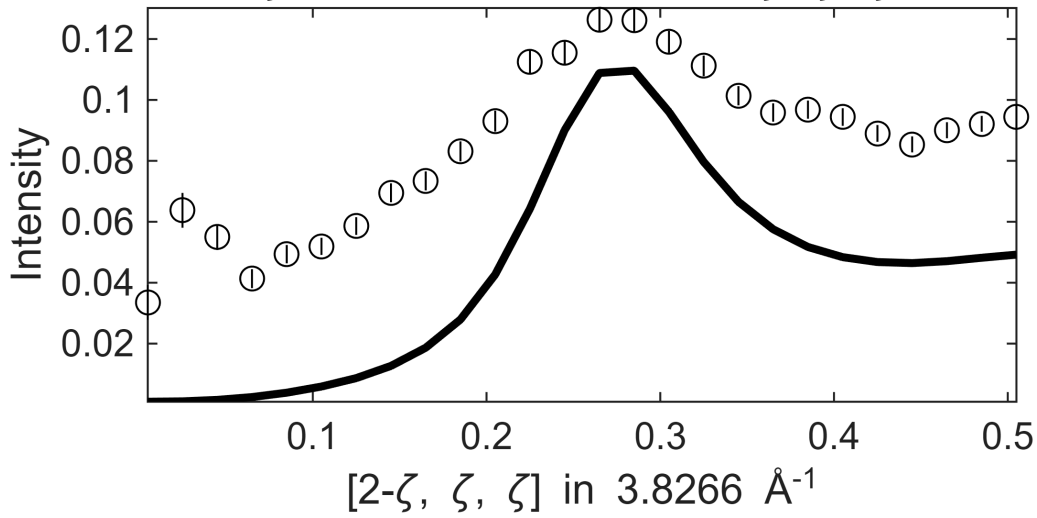
Fe_ei800, w2_800MinFF.sqw

$S=0.62\pm0.024$; $J0= 39\pm 0.2$; $\gamma=58.4824\pm1.24686$
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

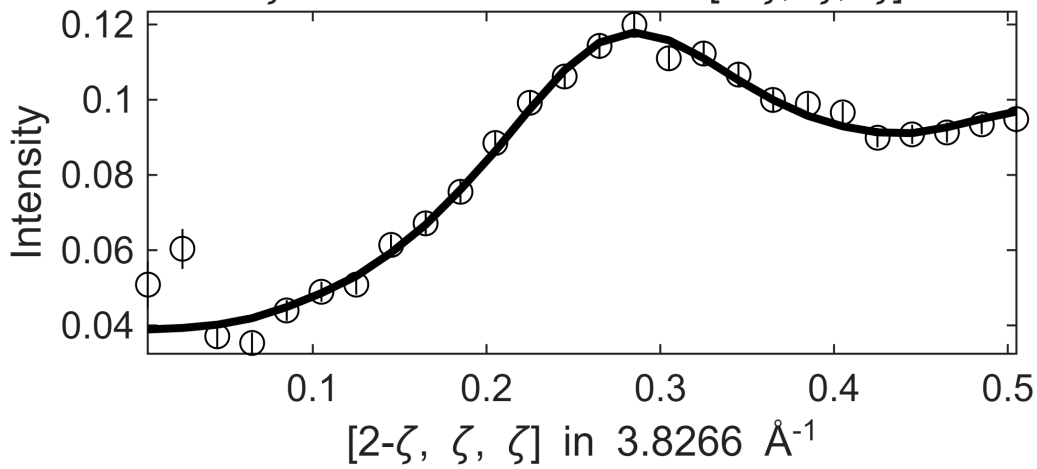
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 235±5 going to 295

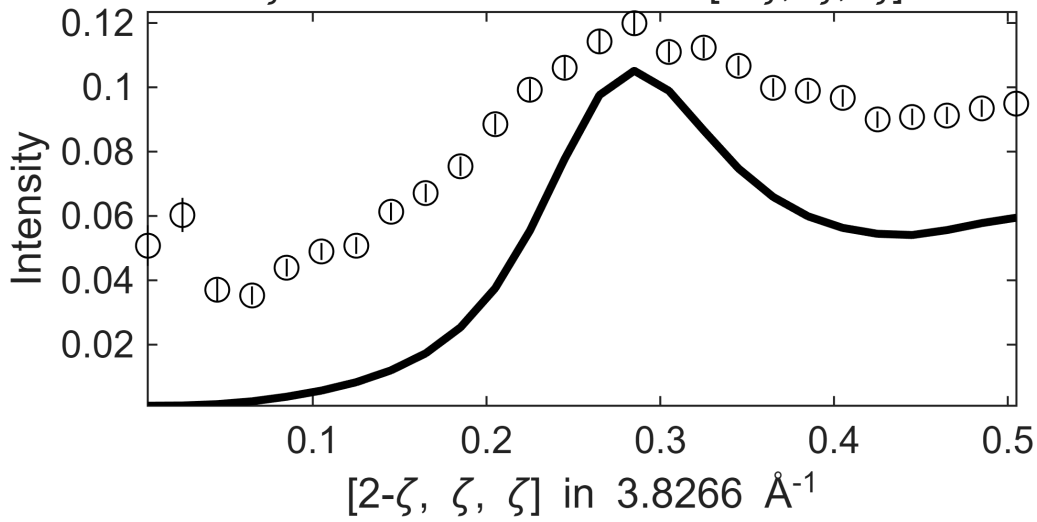
Fe_ei800, w2_800MinFF.sqw

S=0.62±0.056; J0= 39±0.63; gamma=63.4932±7.18872
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

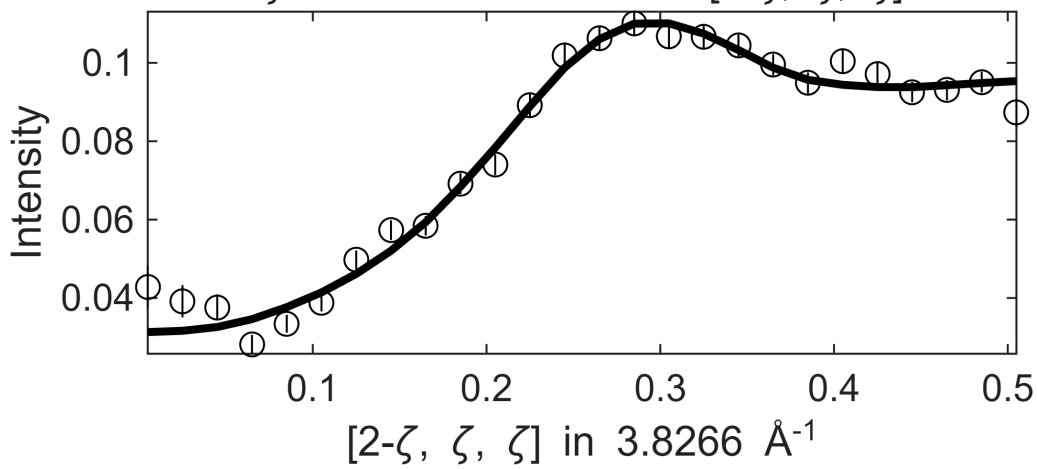
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 245±5 going to 295

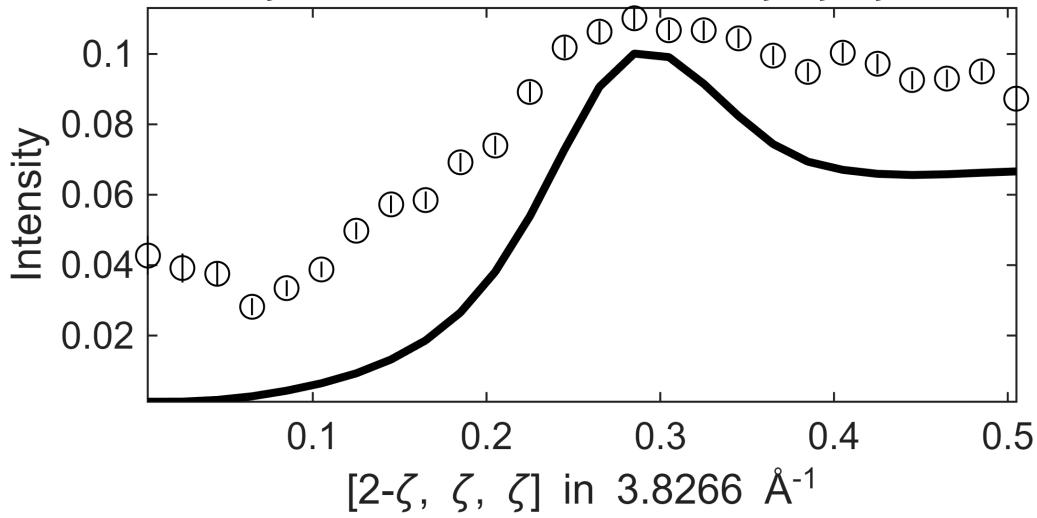
Fe_ei800, w2_800MinFF.sqw

S= 0.7 ± 0.07 ; J0= 40 ± 0.73 ; gamma= 76.3445 ± 9.41151
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

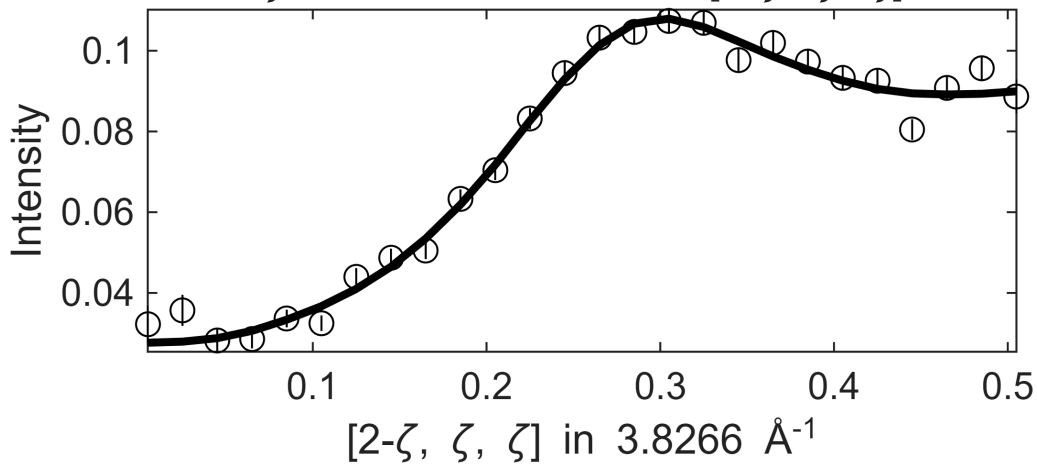
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 255±5 going to 295

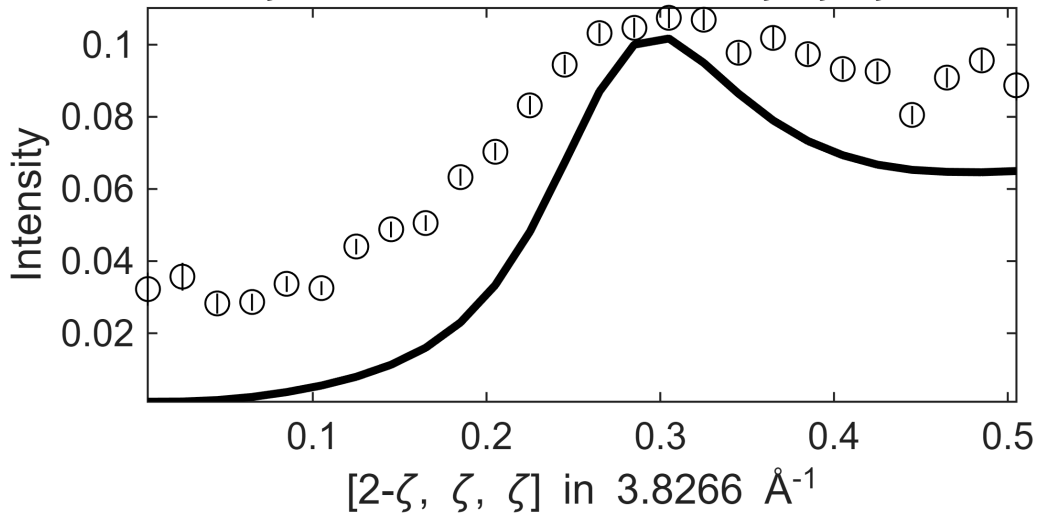
Fe_ei800, w2_800MinFF.sqw

$S=0.67\pm0.026$; $J_0= 40\pm0.24$; $\gamma=71.2267\pm1.80032$
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

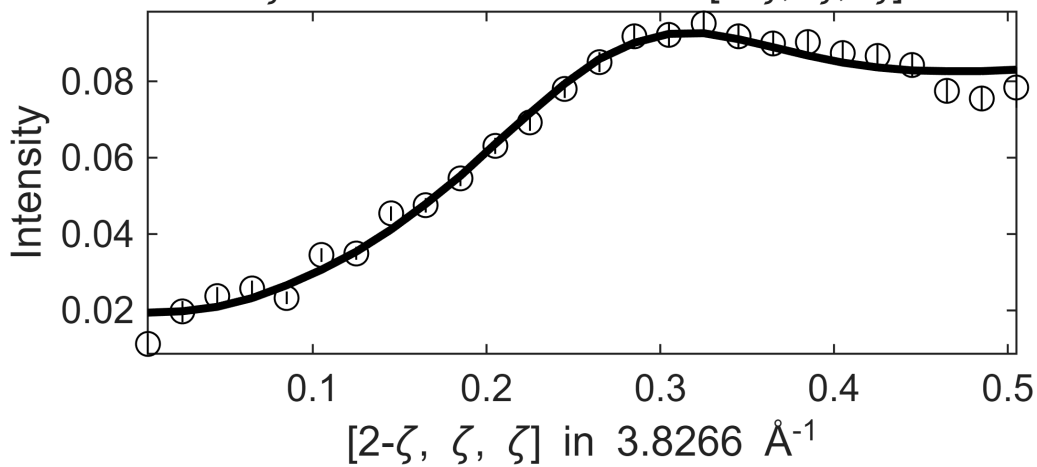
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 265±5 going to 295

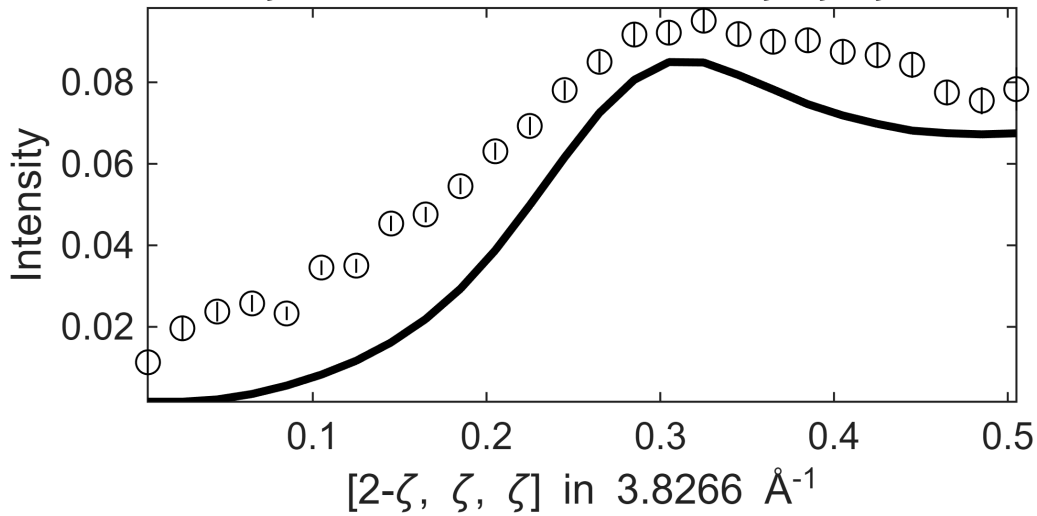
Fe_ei800, w2_800MinFF.sqw

S= 0.9 ± 0.045 ; J0= 43 ± 0.5 ; gamma= 118.567 ± 4.41265
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

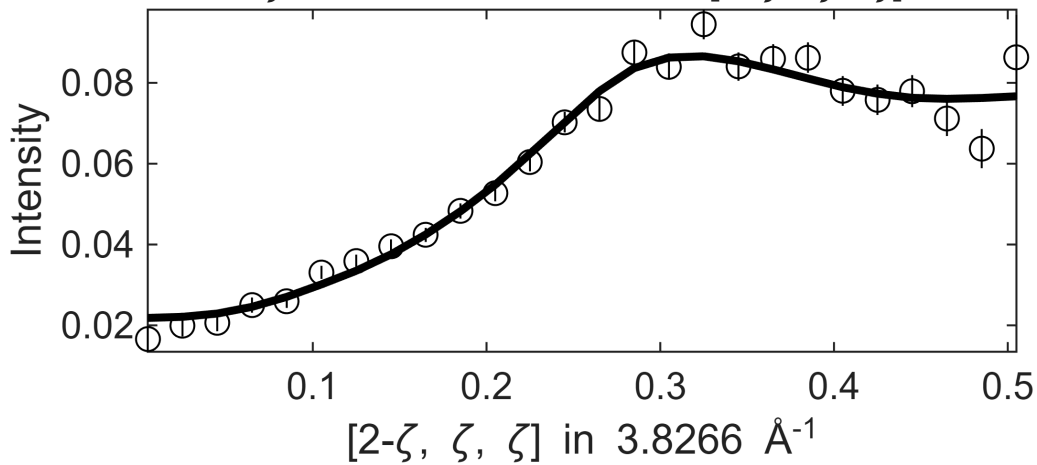
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 275±5 going to 295

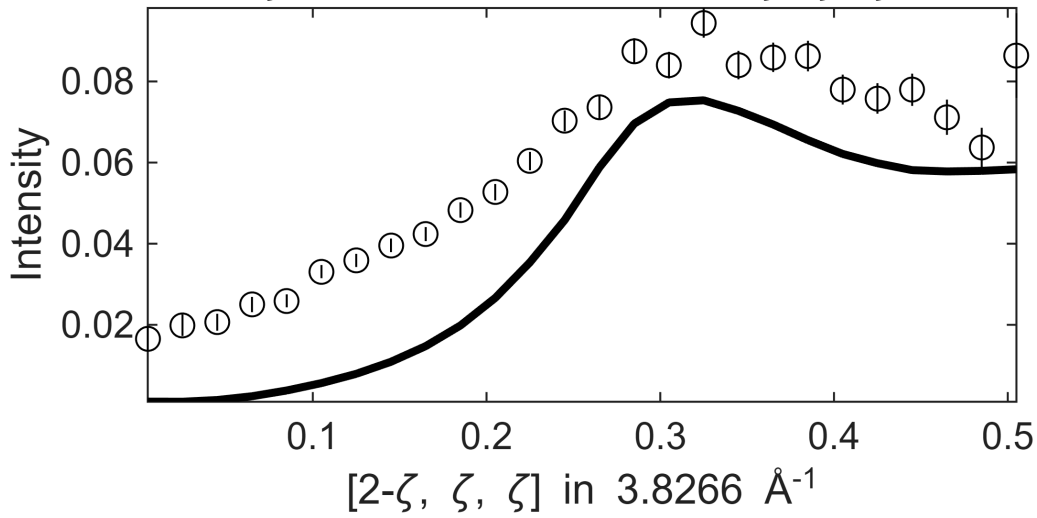
Fe_ei800, w2_800MinFF.sqw

S= 0.7 ± 0.027 ; J0= 42 ± 0.43 ; gamma= 98.6709 ± 2.17674
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

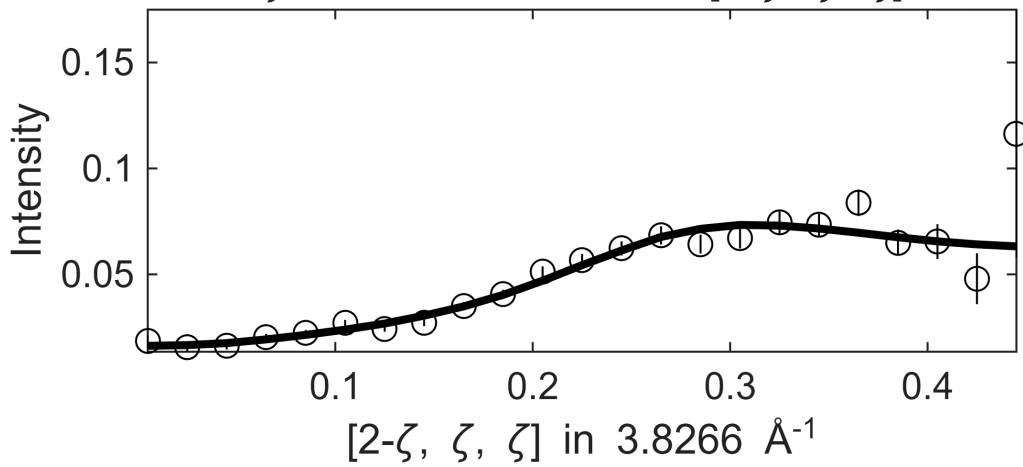
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



 **** En = 285±5 going to 295

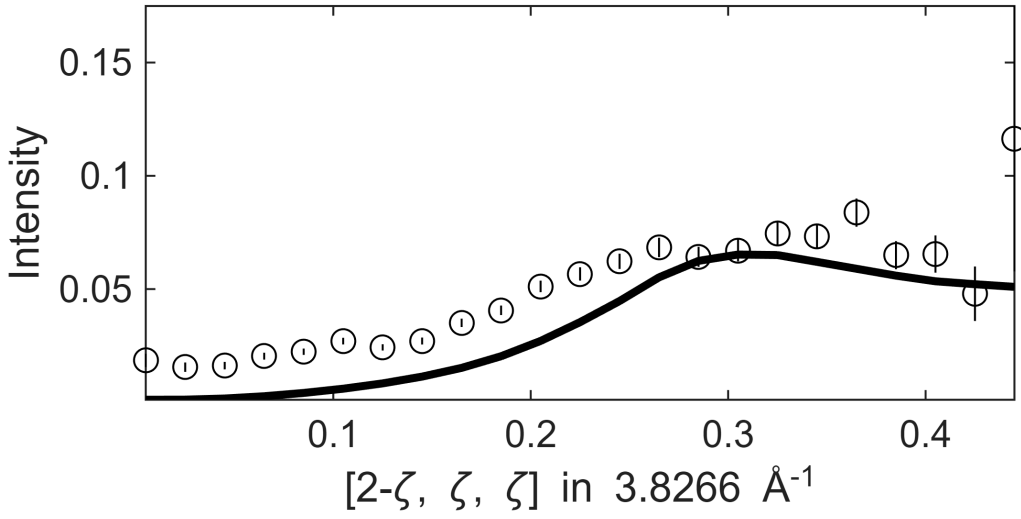
Fe_ei800, w2_800MinFF.sqw

S= 0.8 ± 0.13 ; J0= 45 ± 1.1 ; gamma= 123.381 ± 23.9094
 1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw

1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



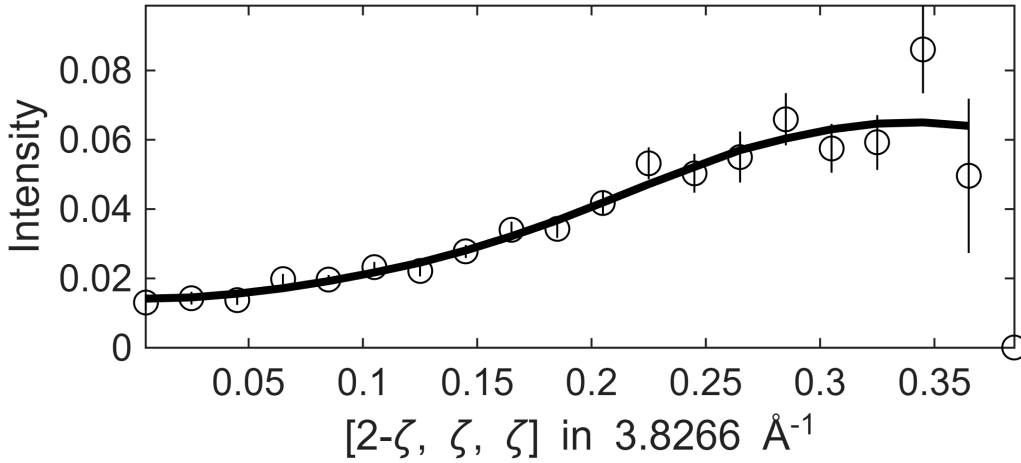
**** En = 295±5 going to 295

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.8462

Fe_ei800, w2_800MinFF.sqw

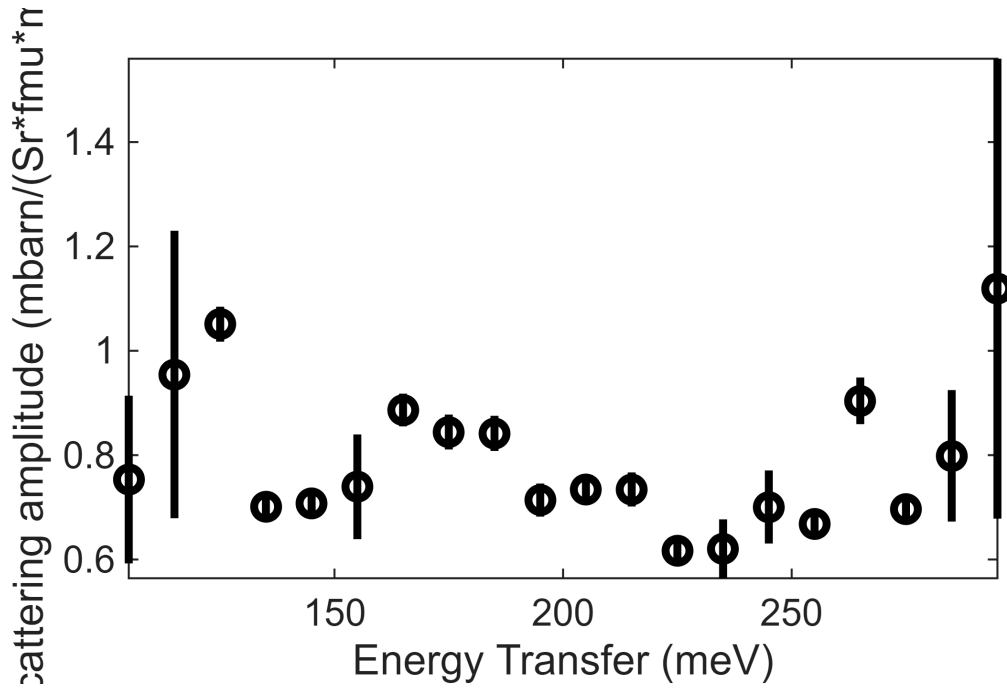
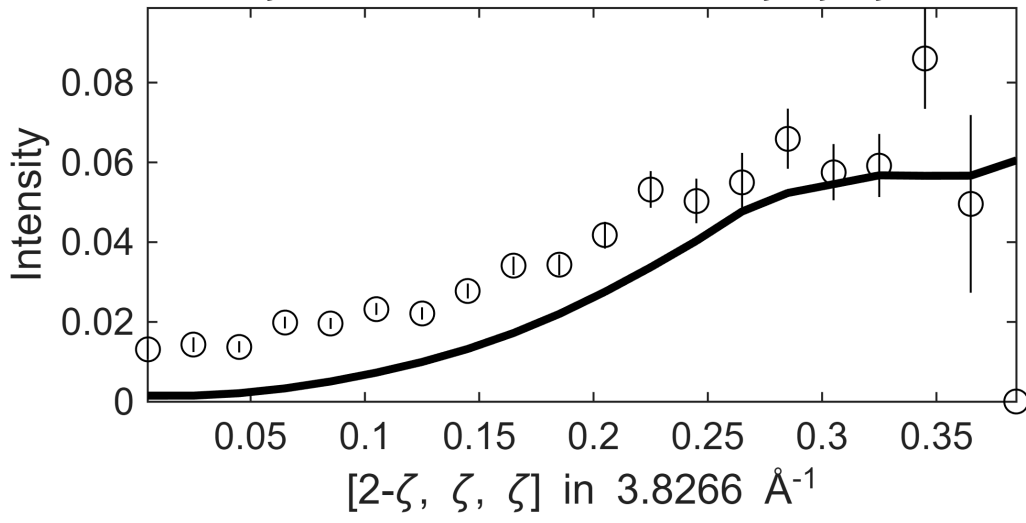
S= 1.1 ± 0.44 ; J0= 45 ± 4.5 ; gamma= 200.794 ± 78.481

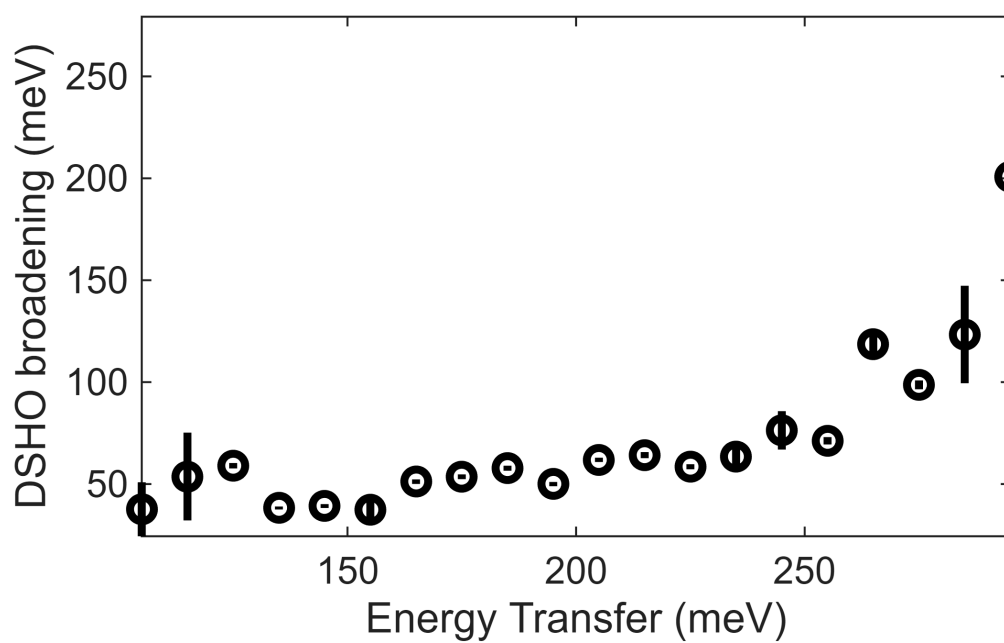
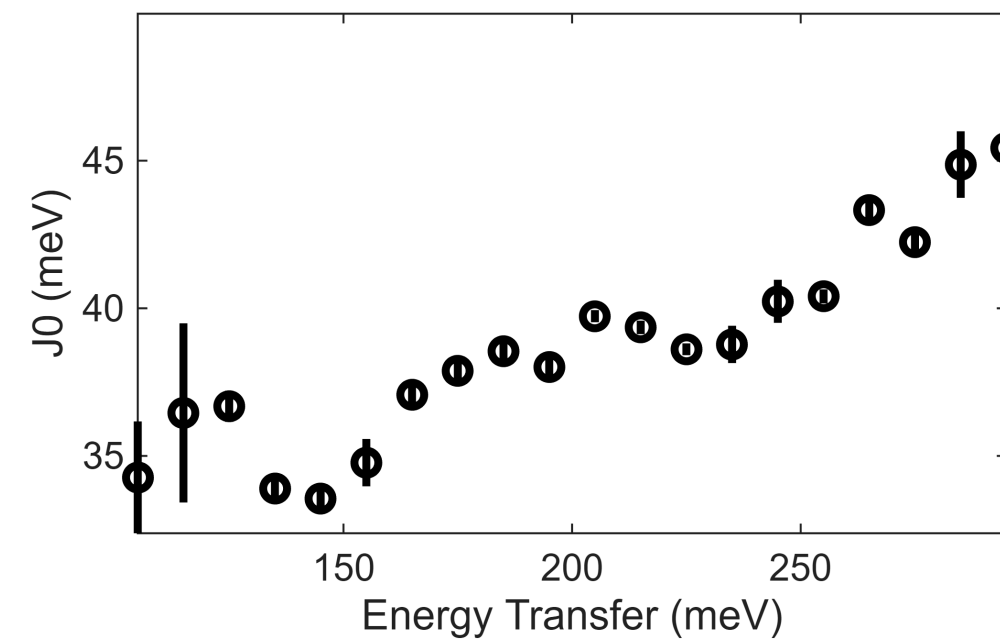
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

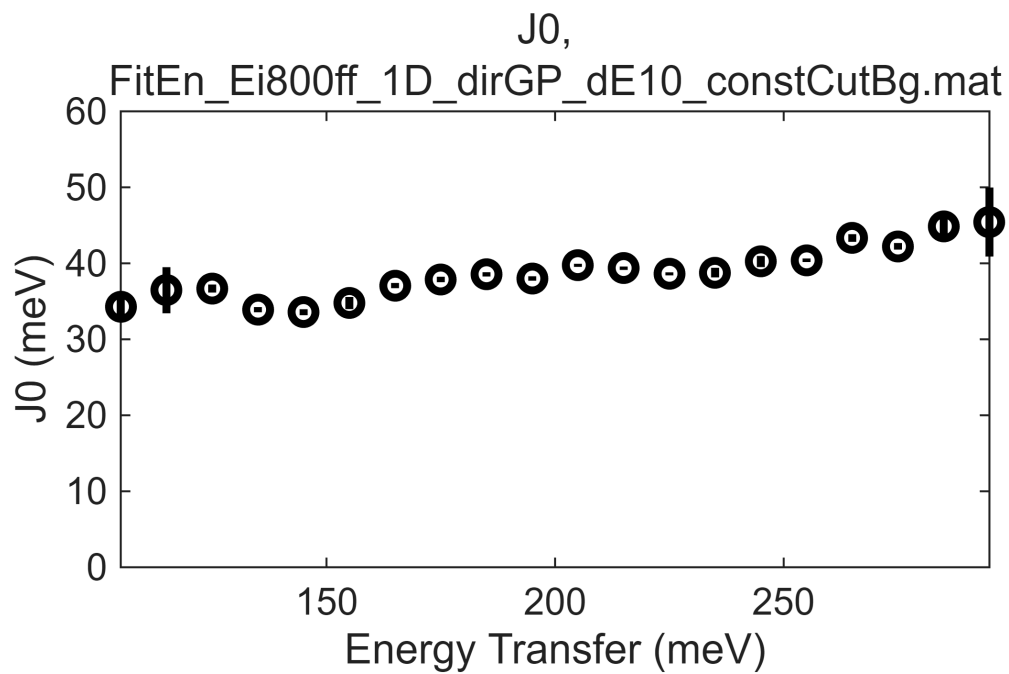
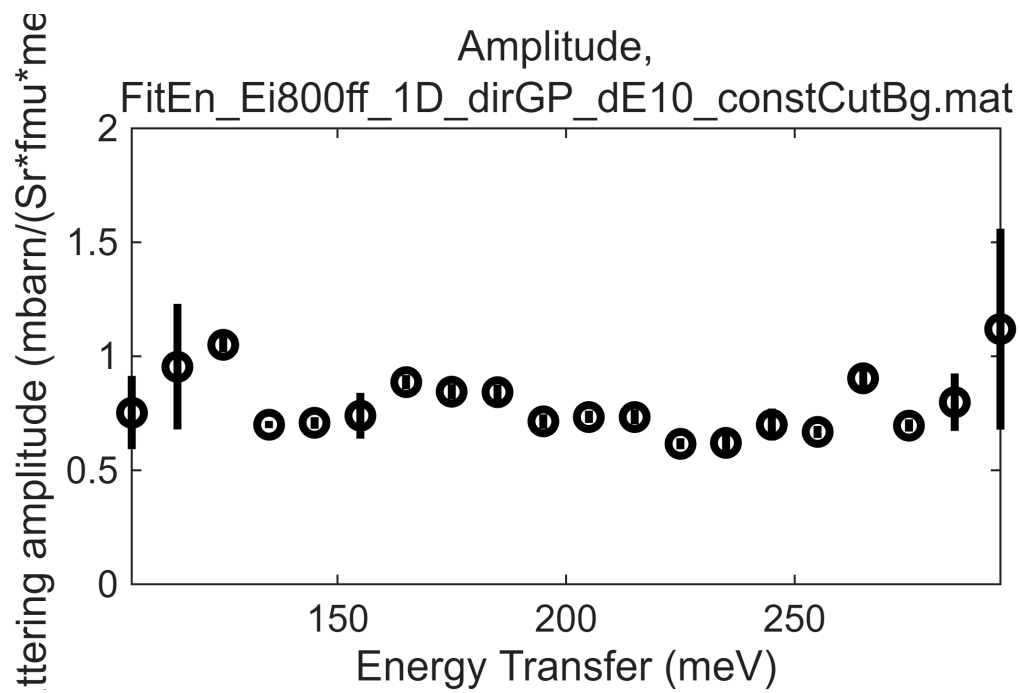


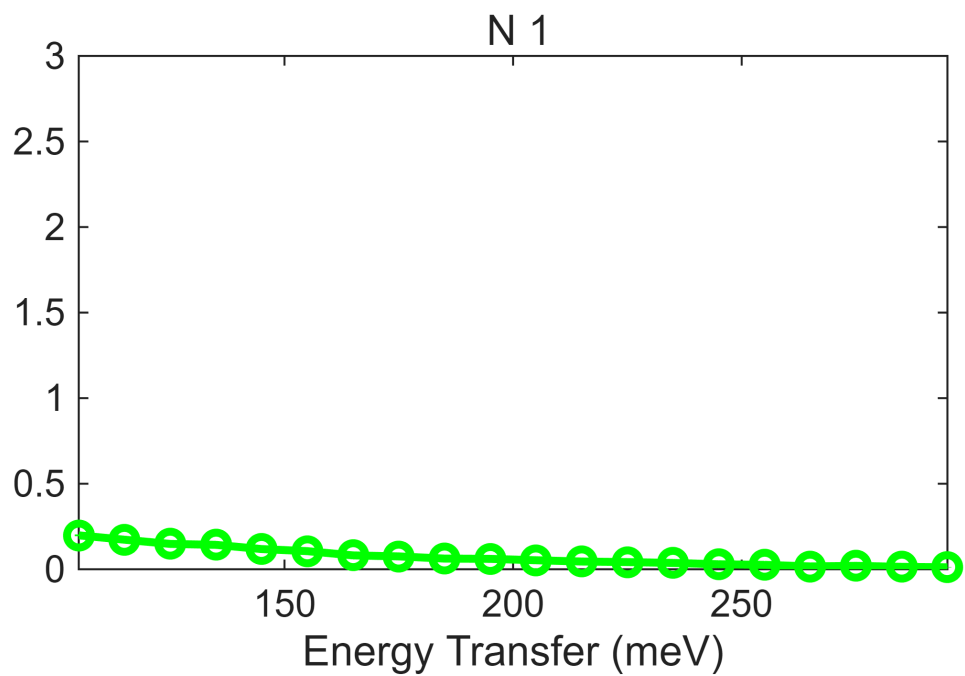
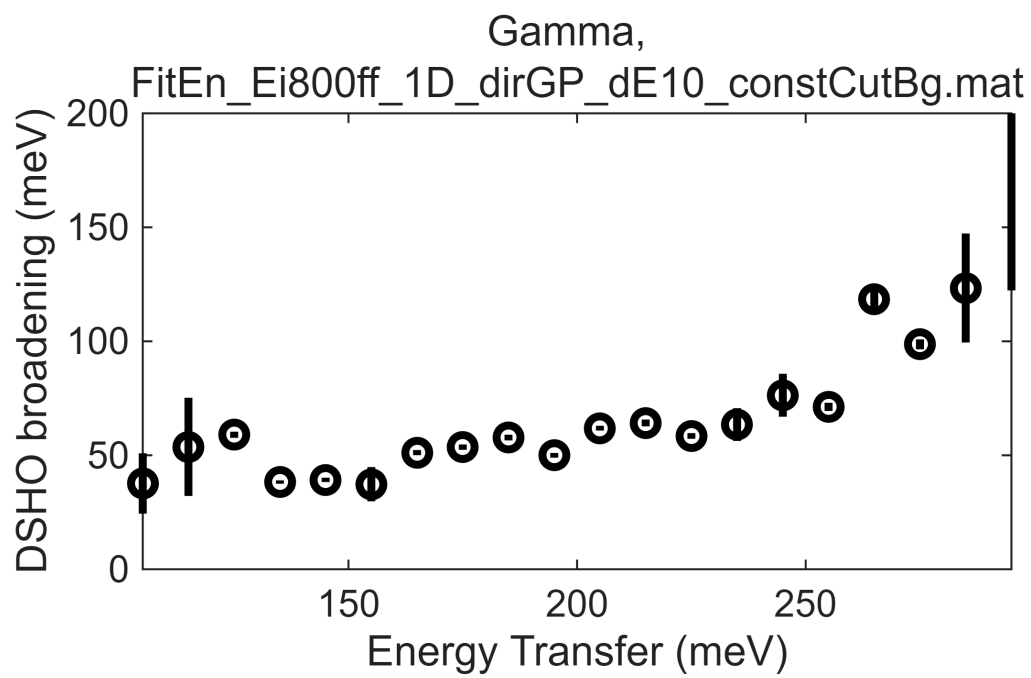
Fe_ei800, w2_800MinFF.sqw

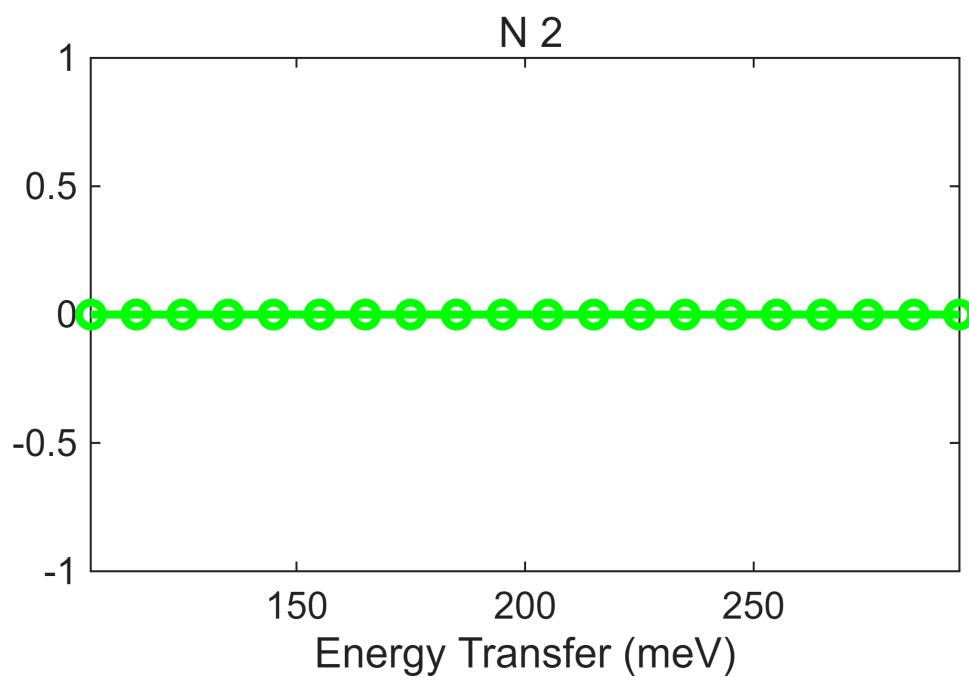
1 in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$,
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$











```
*****
**** CUT: Ei800_off110maxFFGP
*****
*****
**** En = 105±5 going to 295
*****
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